Weather Station and Micro-Climate Sensor Plan

Prairie Plants · An Ecowitt Wi-Fi station with frost, soil, and rain sensors for the orchard, freeze alerts for the red shed and pump house, and an auto-fill plan for the irrigation tank · prices and specifications checked October 18–19, 2026

Companion to the site climate page, which notes that the lot has no thermometer yet, and the water-harvesting plan, which calls for a rain gauge. Prices, specifications, and range figures come from Ecowitt's store pages and manuals and from the other sellers' pages linked below, and they will drift, so follow the links before ordering. This page summarizes them in its own words and links to the sellers; it does not copy their catalogs. Sensor sites are proposals placed with the site master grid and the planting layout, and every distance is measured on that grid.

The Pick in Brief

Buy the **Wittboy GW3001** kit (**$204.99**): a GW3000 gateway and a WS90 outdoor array. It suits a remote, Starlink-connected site better than anything else in Ecowitt's lineup. It costs $5 more than the older GW2001 kit ($199.99), whose gateway lacks the SD slot.

- **Gateway (GW3000).** Wi-Fi and Ethernet, a microSD slot that logs readings as CSV files you can download from its web page or API, a replaceable external radio antenna, and USB-C power. The GW2000 in the GW2001 kit lists no SD slot. A local copy means a Starlink or Wi-Fi outage does not leave a hole in the frost record.
- **Array (WS90).** Wind speed, wind direction, rain, temperature, humidity, light, and UV in one solar-powered unit with no moving parts.
- **Room to grow.** One gateway takes up to 8 temperature/humidity sensors, 8 temperature probes, 8 leaf-wetness, 16 soil, 4 leak, 4 laser-distance, and 4 PM2.5 sensors, a lightning sensor, and 16 water timers, valves, or smart plugs.
- **Pricier units add a screen or a modem, not capability.** The Wittboy Pro HP2564 ($299.99) is the same WS90 with a 7-inch touchscreen console and a shorter add-on list (no irrigation timers). The WittField WS6210 (from $329.99) is a 4G and Wi-Fi receiver for sites without Wi-Fi and needs its own sensors; Starlink makes it unnecessary.
- **Nothing to port-forward.** The gateway sends its readings out to ecowitt.net, so Starlink's lack of inbound connections does not matter.

Reading the Ecowitt Product Names

Ecowitt's store sorts them into Starter Product (complete stations), Console / Gateway, Sensor, and IoT Actuator. The letters in a name tell you which kind of thing it is.

Name	What it is	Examples
GW + 4 digits	Gateway: the radio receiver and the Wi-Fi or Ethernet link. No screen, and no outdoor sensors of its own, only indoor temperature, humidity, and pressure. A kit bundles an outdoor array.	GW3000 alone ($54.99); GW3001 = GW3000 + WS90; GW3002 = GW3000 + WS69; GW2001 = GW2000 + WS90.
WS, **HP**, or **WN** + 4 digits	Console: a gateway with a display screen, often sold in a kit with an array.	WS3900, HP2564 (HP2560 console + WS90).
WS + 2 digits	Outdoor sensor array. This is where wind is measured.	WS90: wind, rain, temperature, humidity, light, UV ($149.99). WS85: wind and rain only ($99.99). WS80: wind, temperature, humidity, light, UV, no rain ($109.99); its page says there is no 915 MHz version for sale in the U.S. WS69: cups, vane, and tipping bucket ($63.99). WS68: wind plus light and UV ($53.99).
WH or **WN** + 2 digits	Add-on sensors that report to the gateway by radio.	WH52 soil, WN31 temperature and humidity, WH40H rain gauge, WN35 leaf wetness.

How It Measures Wind and Direction

The gateway has no wind sensor. It receives readings by radio from the array outdoors (915 MHz, about every 9 seconds) and passes them along: **WS90 → GW3000 → Wi-Fi or Ethernet → Starlink → Ecowitt app and website**, with a copy on the SD card.

- **No cups, no vane.** The WS90's ultrasonic anemometer sends sound pulses between transducers. Wind speeds up or slows the pulses, and the electronics work out speed and direction from the timing.
- **What it reports.** 0 to 89 mph (40 m/s), within ±2 mph below 22 mph and ±10% above, starting at about 1 mph; direction within ±15° in 1° steps; gust is the highest speed in the last 28 seconds.
- **Aim it.** Point the array's NORTH arrow due north (WS90 manual), keep the pole vertical and the array level, and check the arrow again as the last step before tightening the bolts. The fences run within 0.2° of true north, so sight along the west fence; a phone compass reads magnetic north, which is well off true north here.
- **Winter.** Use lithium AAs; the manual warns against alkaline cells, especially with the heater on. The optional 12 V heater cord (10 or 20 m) warms only the wind-measurement area. The rain sensor on top is not heated, and the pole must be within reach of 12 V power.
- **Cups and a vane instead.** The GW3002 kit ($118.99) pairs the GW3000 with the WS69: moving parts, no heater cord, but a listed ±0.3 °C temperature accuracy against ±1 °C for the WS90. The WS90 is still the better fit here.

Shopping List: Buy Now

Choose the **915 MHz (US)** option on every Ecowitt page; sensors and gateway must match. Ecowitt prices are from its store on the dates above. The Amazon links are searches, not specific listings. Ecowitt also sells through its Amazon storefront, but Amazon's prices were not checked.

Item	Qty	Each	Use	Link
Wittboy GW3001 (GW3000 gateway + WS90)	1	$204.99	Base station for wind, rain, temperature, humidity, sun, and UV. The gateway goes in the red shed.	shop.ecowitt.com/products/gw3001
WN31 temperature/humidity sensor	6	$10.99	The outdoor frost network, at sites 1 to 6 below. It measures to −40 °F, but its own screen blanks below 14 °F.	shop.ecowitt.com/products/wn31
Rain Shield for WN31	6	$18.99	One per WN31. The WN31 listing gives no waterproof rating, and this is Ecowitt's weather cover. It fits a 1 to 1.5 in pole.	shop.ecowitt.com/products/shelter
WN31 for the buildings	2	$10.99	Indoor freeze sensors: one in the red shed, one in the pump house (S and P below). No shield needed. With sites 1 to 6 they fill the gateway's 8 WN31 channels, so site 7 goes on the second gateway.	shop.ecowitt.com/products/wn31
WH52 soil moisture, temperature, and EC	4	$27.99	Tree basins at sites 1 to 4. IP66, 1 AA. The WH51 ($17.99) measures moisture only.	shop.ecowitt.com/products/wh52
WH40H rain gauge	1	$39.99	Rain totals beside the WS90; Ecowitt recommends the pairing for accurate rain. Rain only, no snow.	shop.ecowitt.com/products/wh40h
CoCoRaHS gauge (4 in rain and snow)	1	about $42–48	Manual reference for rain and snow, and the official gauge for the CoCoRaHS volunteer network. Ships with a post bracket.	weatheryourway.com/products/official-cocorahs-gauge
microSD card, 32 GB, high-endurance	1	—	Local logging. The gateway reads FAT32 only; larger cards need reformatting.	Amazon search
Lithium AA batteries	24+	—	WS90 2, each WN31 2, WH52 and WH40H 1 each. Lithium for your winters, and for the unheated buildings.	Amazon search
1 in OD aluminum tube, about 6 ft, and **pipe clamps**	1	—	WS90 pole. Its mounting hole fits a 1.0 in pole.	Amazon search: tube, clamps
1 in PVC pipe (1.315 in OD) or **1 in EMT conduit** (1.163 in OD), 5 to 6 ft	6	—	Poles for the shields; their mounting hole is 1 to 1.5 in.	Hardware store

> The Ecowitt items total **$558.80**, or about **$606.80** with the CoCoRaHS gauge. Allow roughly $60 more for the card, batteries, and poles (an estimate, not a quote). To spend less, order 4 outdoor WN31s with shields and 2 WH52s and skip the WH40H (the two indoor WN31s stay in): the Ecowitt total drops to **$402.87**, a saving of **$155.93**.

Handy Add-Ons

Item	Each	Use	Link
WN35 leaf wetness sensor	$54.99	Wet hours for scab and fire-blight timing; one or two in a susceptible apple.	shop.ecowitt.com/products/wn35
WH57 lightning detector	$55.99	Storm alerts. Fits the Rain Shield. If it reports false strikes, check the batteries: Ecowitt says lithium AAs can cause that and suggests alkaline or low-voltage rechargeable cells.	shop.ecowitt.com/products/wh57
WH55 leak detector	$15.99	Pump house floor and tank base; loud alarm.	shop.ecowitt.com/products/wh55
LDS01 laser distance sensor	from $99.99	Tank level or snow depth, 4 cm to 4 m. Buy the single unit; the $129.99 kit adds a GW1200 gateway you will not need. Rated −4 to 122 °F, so it will miss your coldest nights. 8 AA.	shop.ecowitt.com/products/lds01
WH41 PM2.5 air-quality sensor, solar	$71.99	Wildfire smoke.	shop.ecowitt.com/products/wh41
WFC01 WittFlow water timer	from $59.99	Timer with a built-in flow meter and water temperature, for one zone.	shop.ecowitt.com/products/wittflow-wfc01
WFC02 smart ball valve	from $119.99	Motorized brass ball valve: 1 in with flow meter $129.99, 1 in without $119.99, or 2 in $139.99 (the meter is 1 in only). BSP threads need NPT adapters. It works above 32 °F only: drain it for winter. 8 AA.	shop.ecowitt.com/products/wfc02
Tank auto-fill: LDS01 + WFC02	$229.98	Valve on the tank's supply line and laser in the lid, both on the gateway's 915 MHz radio rather than Wi-Fi. Whether the laser can stop the valve is unconfirmed: see Irrigation Tank: Auto-Fill and Level Management for the plans and the question to ask Ecowitt first.	WFC02 manual, LDS01 manual
AC1100 smart plug	$29.99	Switches a heater, pump, or fan on a schedule or from a sensor reading, and shows live watts. 12 A maximum (1,440 W). For a heater in each building, see Freeze protection below.	shop.ecowitt.com/products/wittswitch-ac1100
12 V heater cord, 10 or 20 m	see page	Clears snow and ice from the WS90's wind sensor. The pole must be near 12 V power.	shop.ecowitt.com/products/extension-cord
GW3000 second gateway	$54.99	For the north rows and the northwest frost pocket; it needs power and Wi-Fi there. Add a WN31 ($10.99) and shield ($18.99) for the pocket, since the first gateway's 8 WN31 channels will be full.	shop.ecowitt.com/products/gw3000

Freeze protection with the AC1100 (red shed and pump house)

Yes. In the Ecowitt app, a Smart rule can use a WN31 channel's temperature as its trigger and switch the plug on or off, so the indoor sensors (S and P) can run a small heater or warming pad. The rules run in the gateway, so a Starlink outage does not matter. The sensor and the plug must be on the same gateway (the AC1100 manual says rules cannot yet use another gateway's sensors); here that is the GW3000 in the shed. Two plugs cost $59.98.

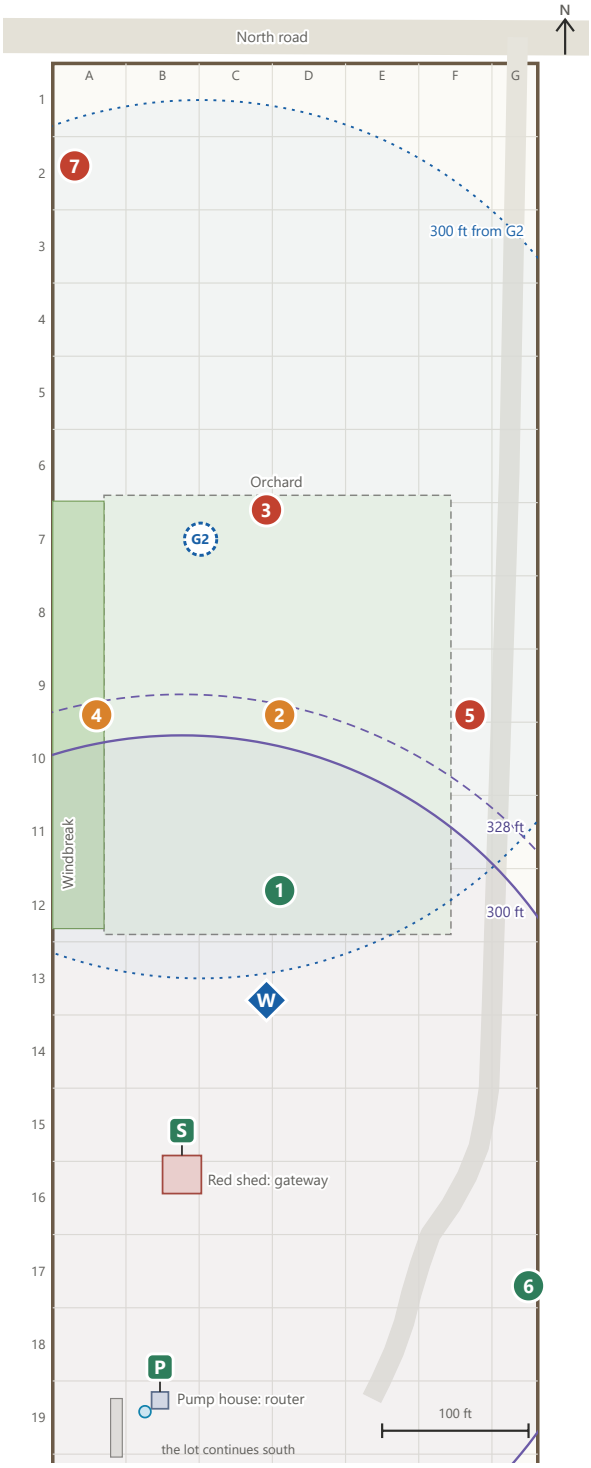
Two rules per building, plus a warning (in the plug's Smart screen; effective period all day)

Step	Trigger	Action
Heat on	That building's WN31 temperature is less than 38 °F	AC Switch: On
Heat off	That building's WN31 temperature is greater than 45 °F	AC Switch: Off
Warning	The same sensor falls below 35 °F (an ecowitt.net alert)	E-mail: the heat is not keeping up

- **Settings.** The app lists each WN31 channel as "Temp and Humidity CH_: Temperature," with "is less than" and "is greater than" tests. The WN31 reads within about ±2 °F every 61 seconds, so 38 °F leaves margin above freezing and the 7 °F gap prevents cycling.
- **Rating.** The plug is rated 12 A (1,440 W at 120 V) and cuts the load off itself only at 18 A, so a 1,500 W space heater (12.5 A) is over the rating and nothing would stop it. Keep the load under about 1,000 W, plugged straight into the AC1100 and then into the wall outlet (GFCI in the pump house), with no extension cord.
- **Choose a heater that restarts and has its own thermostat.** A heater or pad with a knob thermostat heats as soon as power returns and holds its own set point (about 40 °F) if the radio or gateway fails; many digital or remote-control heaters and electric blankets stay off after a power cut. Each thermostat covers the other: if the heater's sticks on, the rule cuts the plug above 45 °F.
- **Where it can fail.** The chain is WN31, radio, gateway, radio, plug. Ecowitt's "communication unstable" alert disables Smart rules, and the shed gateway is rated only to 14 °F, so do not rely on the plug alone for the well pump. The 35 °F alert warns you if the heat did not keep up.
- **Energy.** If the pump house runs from the battery bank, a 100 W pad on for half of a 14-hour night uses about 0.7 kWh, and a 750 W heater on the same schedule about 5 kWh. Warm the pump or pressure tank directly rather than the room.
- **Not published.** Ecowitt gives no operating-temperature or weatherproof rating for the AC1100 and does not say what state it takes after a power cut. Mount it dry and warm, and test both before winter; the app shows its signal and live watts.

Where the Sensors Go

Proposed sites, to scale with north up and the 50 ft grid. The gateway goes in the red shed, where there is power. S and P are the indoor freeze sensors, inside the shed and the pump house. Distances are straight lines from the shed's center. Ecowitt lists 300 ft (100 m) of open-air range for the WN31 and WH52 and about 328 ft (100 m) for the GW3000's receiver; the WS90 transmits farther (500 ft). Flat, quiet ground like yours may do better, so treat the colors as a list of what to test, not a verdict.



Purple rings: 300 and 328 ft from the shed. Dotted blue ring: 300 ft from G2.

#	Site	SiteRef	Elev., ft	From shed	Sensors
1	South orchard beside tree R2-C7	D12-E5-S15	4,754	205 ft within range	WN31, WH52
2	Mid orchard beside tree R8-C7	D9-E5-S45	4,750	321 ft at the limit	WN31, WH52
3	North orchard beside tree R15-C7	C7-E46-S5	4,744	458 ft beyond range	WN31, WH52
4	West edge between the windbreak and column C1, level with R8	A9-E30-S45	4,748	319 ft at the limit	WN31, WH52
5	East edge east of column C13, level with R8	F9-E35-S45	4,752	371 ft beyond range	WN31
6	High point inside the east fence	G17-E25-S35	4,762	249 ft within range	WN31
7	Northwest frost pocket low corner by the road	A2-E15-S20	4,740	693 ft beyond range	WN31 (second gateway)
W	WS90 pole open ground south of the orchard	C13-E46-S40	4,755	132 ft within range	WS90, WH40H, CoCoRaHS gauge
G	Gateway red shed, power here	B16-E38.25-S9	4,756	—	GW3000 (in the kit)
S	Red shed, inside inside wall, a couple of feet off the floor	B16-E38.25-S9	4,756	—	WN31 (indoor)
P	Pump house, inside inside wall, near the water pipes	B19-E23.25-S13.25	4,759	155 ft within range	WN31 (indoor)
G2	Second gateway, if needed near tree R14-C4	C7-E1-S25	4,745	434 ft	needs power and Wi-Fi: 589 ft from the router

- **Cold air.** The northwest pocket (7) is the lowest ground and the high point (6) the highest. The climate page expects cold air to drain to the pocket, so the gap between the two is the number to learn.
- **Freeze check in the buildings.** Hang each indoor WN31 (S, P) on an inside wall a couple of feet off the floor, near the water pipes and away from the inverters and any heater, and name the channels in the app. The WN31 listing says ecowitt.net can e-mail alerts for its sensors, and Ecowitt's announcement says the e-mails come from alert@notifications.ecowitt.net, so add that address to your safe senders. The sensor is good to ±1 °C (about ±2 °F), so set the low alert near 35 °F to be warned before 32 °F, and test it once with the sensor in a bag of ice. Test P's signal first, since it passes through two buildings' walls.
- **Test the range first.** Before burying anything, run each WN31 at its site for a day and read its signal and battery on the gateway's Sensors ID page (GW3000 manual). The WS90 manual also says to run the WS90 for a week at a temporary spot before mounting it for good.
- **Second gateway.** If sites 3, 5, and 7 fail the test, a second GW3000 at G2 would be within 300 ft of sites 1, 2, 3, 4, 5, and 7. It needs power and Wi-Fi there, and G2 is about 589 ft from the pump house, where the Starlink router is.

Irrigation Tank: Auto-Fill and Level Management

The site data lists a 3,000 gal black poly tank beside the pump house, about 8 ft across and 8 ft tall, so each inch of depth holds about 31 gal. A straight-sided tank that size holds about 3,008 gal brim-full, so the level at the overflow fitting is a little lower: measure it. The site data does not say how the tank is filled today (the pump, the pipe size, the inlet and overflow heights); check those before buying. The LDS01 laser and the WFC02 valve reach the gateway by 915 MHz radio, not Wi-Fi. The tank is about 164 ft from the shed, inside the 328 ft rating, but it sits beside the pump house and the steel solar container, so test the signal at the valve's real spot.

Your idea works on paper: **a daily timer opens the valve, the laser stops it when the tank is full, and the same rule shuts it again if the tank is still full the next day.** The timer and a stop-on-sensor rule are documented. What Ecowitt does not say is whether the laser's level can be the trigger that stops the valve, so there is a plan for each answer.

What Ecowitt's manuals confirm, and what they leave open

The idea	What the manual says	Source
A daily timer that keeps running if the gateway link drops	**Confirmed.** A Plan is saved on the valve itself.	WFC02 manual §5.1.2.3
A rule that closes the valve while the timer is running	**Confirmed,** with a temperature rule as the example: the valve shuts at once, and the Plan resumes at its next scheduled time.	WFC02 manual §5.2
Rules that work with Starlink down	**Confirmed.** Once downloaded, the gateway runs them on its own sensor data and logs to Ecowitt later.	WFC02 manual §5.1.2.4
The laser's level as the rule's trigger	**Not stated.** The manual says only "supported parameters in the gateway," and its example trigger list shows no laser. The LDS01 manual says only that it works with a "smart plug, water timer, and leak sensors."	WFC02 manual Fig. 22; LDS01 manual §7
The rule firing again while the tank stays full	**Not stated.** The manual's sample log shows one always-true rule logged three times within 20 minutes (18:01, 18:02, 18:20), a hint that it can.	WFC02 manual Fig. 26
The laser's own low and high "alerts"	**Notifications.** The manual describes alert notifications and does not tie them to the valve, so the valve's rule has to read the level itself.	LDS01 manual §5.1.5

Plan A: the laser stops the valve

If the app offers the laser as a trigger

1. **Timer.** In the valve's Plan screen, open it at 100% every day, at a time the well pump can run, for a capped time (see Values to set).
2. **Stop rule.** In Smart, add the trigger "laser depth is at least FULL" (or "air gap is at most …," whichever the app lists) with the action Water Valve: Off. Set the effective period to all day, every day.
3. **Still full the next day.** The timer opens the valve and the rule should shut it again within about 90 seconds (one laser report, about 80 seconds, plus the 7 seconds the valve takes to close), costing up to about 19 gal at 13 gpm.
4. **Prove it once.** With the tank at FULL, start a 3-minute manual run (Duration) in the app. If the valve shuts within about 90 seconds, the rule fires again and Plan A works; the Smart log lists each firing. If it runs the full 3 minutes, the rule fires only when the level crosses FULL, so use Plan B's cap.

Plan B: timer and meter only

Works whatever the app allows

1. **Timer by volume.** The 1 in valve with the flow meter can run a Plan by volume: your daily use plus about 20%, with a time cap as the second limit. The manual's screens enter volume in liters (1 gal = 3.8 L).
2. **Laser as the gauge.** Its low and high alerts tell you when the numbers are wrong, and the valve's log reports each day's gallons, so retune the volume monthly.
3. **Float valve as the stop.** A mechanical float valve on the inlet is the final shut-off whatever the electronics do.
4. **A script is a project, not a plan.** A program on your network could read the laser and command the valve, but Ecowitt's local API document (August 2024) covers only the WFC01 and AC1100.

Where you manage it

Ecowitt app (phone)	The main control: pair the devices, open or close the valve, edit the timer and Smart rules, read the valve log (runs, gallons, alerts) and the Smart log, and set the laser's Total Height, calibration, and level alerts. Timers and rules are set through Ecowitt's server, then run in the gateway.
Gateway web page (the GW3000's address)	The laser's Total Height, offset, and data filter, as an alternative to the app, and the SD card files. The manual puts the level alerts in the app only.
ecowitt.net (browser)	Level history graphs, and e-mail alerts for sensors (check that the laser appears in its alert list). I found no valve controls there; the manuals describe valve control only in the app.
Alerts to expect	Valve: no water, leak after closing, communication unstable (Smart rules disabled), control failed, low battery, valve abnormal, flow meter disconnected. Laser: low and high level.

Values to set (measure the real tank first)

Setting	Where	Value and why
Total Height	Laser: app or gateway web page	Tape-measure from the laser's lower edge to the tank floor (close to the tank's 8 ft). The laser then reports Depth = Total Height − Air. Left unset, it shows only the air gap.
Overflow depth	Measure once	Fill to the overflow fitting and note the Depth reading. Every level below keys off it.
FULL	Smart rule and the laser's high alert	Overflow depth minus 4 in, which leaves about 125 gal of headroom for the laser's reporting delay and the valve's closing time.
LOW	The laser's low alert	At least 12 in of water (about 376 gal): Ecowitt advises more than 0.3 m, and under 8 in (about 251 gal) the beam may read the tank floor. Raise it if the irrigation pump needs more.
Timer start	Valve: Plan	Daily, when the well pump can run, such as midday if it runs off the solar system.
Timer cap	Valve: Plan duration, and volume with the meter	Minutes = gallons to add ÷ gpm; 1,000 gal at 8 gpm takes about 2 hours. Time the first fill, then adjust. Keep it tight under Plan B and generous under Plan A. Give every open command a duration or volume, never "always on."
Opening	Valve	100%. The meter reads 1 to 13 gpm, so if the pump delivers more, step the opening down in 10% steps.
Data filter	Laser calibration	If the reading jumps while water runs in, raise the Data Filter Factor (0 to 4; 2 uses the median of the last 5 readings).

Install and protect

- **Laser.** A 45 to 50 mm hole in the lid on a level surface (a flat hatch or a level plate, if the lid is domed), the bubble level centered, 40 mm of clear space in front of the lens, and at least 3.4 in from the tank wall (Ecowitt's rule is 3.5 cm per 100 cm of height). Keep the beam clear of the inlet stream, pipes, and any float arm. Ecowitt notes extra jitter when the air gap is about 1.1 to 1.5 m, which centering the bubble usually cures. The display and battery box connect by 1.5 m cables, so they need not sit on the lid; the sensor is rated to 122 °F and a black tank in summer sun can run hotter than the air.
- **Valve.** On the supply line with the meter's arrow along the flow, thread tape on every joint, and the actuator upright. End the inlet above the water line (an air gap, so the tank cannot siphon back) and fit an overflow at least as large as the inlet, aimed away from the pump house and containers. If the fill line comes straight off the well pump, check that the pump has pressure control so a closed valve cannot dead-head it.
- **Backstops.** The timer keeps running if the radio link drops, but the stop rule does not: "communication unstable" disables Smart rules. A low battery locks the valve and closes it if it is open. Keep the float valve and overflow as the last line of defense.
- **Keeping it running.** Both devices take 8 AA alkaline cells; change them each spring (the valve lasts over 10 months at one cycle a day). Compare the laser's Depth with a tape reading each month, and watch its heater counter: more than about 30 a day means a dirty or fogging lens. In fall, close the supply, drain the valve and meter (rated above 32 °F), and take the batteries out of both.

Tank parts

Item	Price	Link and note
LDS01 laser distance sensor, single unit, 915 MHz	$99.99	The $129.99 kit adds a gateway you already have.
WFC02 valve, 1 in with flow meter, 915 MHz	$129.99	1 in without the meter $119.99; 2 in without a meter $139.99. Only the 1 in kit can meter volume. The two Ecowitt parts total $229.98.
Thread adapters (BSP to your pipe), not included	—	Ecowitt's picks on its WFC02 page: 1 in universal, 1 in valve, 1 in meter, 2 in valve (Amazon; availability not checked).
16 alkaline AA batteries	—	8 per device. The valve manual says no rechargeables.
Float valve, overflow fitting, thread tape, 1-7/8 in hole saw	—	Hardware store. A 1-7/8 in (48 mm) saw fits the laser's 45 to 50 mm lid hole; a 2 in saw cuts just over it.

> **Settle the open question before spending $230.** Ask support@ecowitt.com: (1) In the Ecowitt app, can the LDS01's depth or air gap be the trigger of a WFC02 Smart rule on a GW3000? (2) If the level is already above the trigger when the timer opens the valve, does the rule close it again, or only when the level crosses the trigger? Or buy both, pair them, and look in the valve's Smart screen on arrival: Ecowitt's own store takes change-of-mind returns only within 14 days of delivery, at your cost (returns policy). Either way, Plan B works.

Before You Order

- **Cold.** The GW3000 is rated down to 14 °F (−10 °C). The coldest nights average −11 °F, and the record is −32 °F (`data/climate.json`, Prairie SNOTEL). Unless the shed stays above 14 °F, keep the gateway in a heated spot or an insulated box with a small thermostat-controlled heater, and run its antenna outside. The freeze alerts for the buildings depend on this gateway, Wi-Fi, and power, and the gateway's own indoor reading stops at 14 °F. An AC1100 smart plug can switch a heater in each building from the indoor sensors' readings. The WS90, WH52, and WN31 transmit down to −40 °F; the LDS01 is rated only to −4 °F.
- **Data.** ecowitt.net keeps day-by-day detail for 3 months, weekly data for 1 year, monthly for 2 years, and yearly for 4. The SD card writes CSV files in its root folder. It must be FAT32 (cards over 32 GB need reformatting with a tool such as Rufus), can be swapped without a restart, and is managed from the gateway's web page (GW3000 manual). Back up the CSV files monthly. The repository only needs the daily lows and highs, kept apart from the SNOTEL record, since on-site readings are not station readings.
- **Setup.** Use two lithium AAs in the WS90, run it for a week at a temporary spot, and aim its north arrow. In the gateway's Device Settings (GW3000 manual), turn on Temperature Compensation to reduce sun heating of the WS90's temperature. Check each temperature sensor in an ice bath (32 °F) before it goes outside, and note any offset.
- **Soil sensors and rain gauges.** The WH52 (sites 1 to 4) goes in the basin of the nearest tree, ideally one used for the perc tests. Put the CoCoRaHS gauge and the WH40H on separate posts a few feet from the WS90, clear of the trees.
- **Accuracy.** The WS90's temperature sensor is not fan-ventilated, so it can read warm at midday in calm sun; that matters less on frost nights. The WN31 and WS90 are both listed at ±1 °C. Neither the WS90's rain sensor nor the WH40H measures snow, so use the CoCoRaHS gauge and the Prairie SNOTEL for winter precipitation.
- **Starlink.** The gateway joins 2.4 GHz Wi-Fi only. If it will not connect, temporarily split the 2.4 and 5 GHz networks in the Starlink app (a third-party how-to), or use the Ethernet port.
- **Wi-Fi at the shed.** The shed is about 155 ft from the pump house. The GW3000 lists 30 m (98 ft) of Wi-Fi range, so check the signal at the shed with a phone, and use the gateway's Ethernet port or a mesh node if it is weak. The GW3000 manual also describes moving the antenna on an extension cable to the best-signal spot; check that the cable is in the box.
- **Still unknown.** Whether the shed and pump house stay above freezing, and the shed above 14 °F (the indoor sensors will show it), whether Wi-Fi reaches the north orchard, how far the radio really carries on this ground and through the pump house walls, how the tank is filled today, and whether the laser's level can stop the valve (the Ecowitt question in the tank section). The range test settles the radio question.

Alternatives Considered

Option	Price	Trade-off
Ecowitt GW1210 (GW1200 gateway + WS90)	$183.99	Same array and about $16 less than the GW2001, but a Wi-Fi-only gateway with no Ethernet port. Price from a price tracker (Amazon), October 8.
Ecowitt GW2001 (GW2000 gateway + WS90)	$199.99	The older gateway lists no SD slot.
Tempest Weather System	$349	Easiest and fully sealed, with lightning detection. No soil, leaf-wetness, or extra temperature add-ons.
Ambient Weather WS-2902	about $200	Popular console station with wind cups and a tipping bucket. No soil sensors, and text alerts need the paid AWN+ plan.
Ambient Weather WS-5000	$449.99	Ultrasonic wind and a color console; about twice the Ecowitt price.
Davis Vantage Vue + WeatherLink Console	$725	The most rugged, with a 1,000 ft radio range. No solar-radiation or UV measurements listed; about 3.5 times the cost.

Sources

- Ecowitt store pages for each item linked above (shop.ecowitt.com): prices, specifications, compatible-sensor lists, and range figures, checked October 18–19, 2026. The GW3001 page gives the GW3000 and WS90 specifications.
- Ecowitt GW3000 manual (SD card, antenna extension, sensor signal page, Temperature Compensation, history retention) and WS90 manual (mounting, north alignment, batteries, one-week trial); the WFC02 manual and LDS01 manual (radio link, valve modes and limits, tank-sensor setup and level alerts); the AC1100 manual (Smart-rule triggers and actions, alerts, ratings).
- Ecowitt help pages: the alert e-mail announcement, the local IoT API document, the returns policy, and the contact page, which lists support@ecowitt.com. WFC02 prices are from its store page (1 in with meter, 1 in without, and 2 in).
- CoCoRaHS gauge: WeatherYourWay. Alternatives: the Tempest, Ambient Weather, and Davis pages linked in the table, and a price tracker (localsmarthomeguide.com) for the GW2001, GW1210, and Tempest.
- Cold record: Prairie SNOTEL, in `data/climate.json` (see the site climate page). Sites, distances, and elevations: `data/site.json`, the planting layout, and the USGS lidar terrain, through the shared grid helpers (see the site master grid and homestead plot map).

Prairie Plants · Weather station and micro-climate sensor plan · created October 19, 2026 · sites, distances, and elevations are calculated from `data/site.json` and the lidar terrain; prices were checked October 18–19, 2026 and will drift.